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Antibiogram, Virulence Genes, and Biofilm-Forming Ability of Clinical *Salmonella enterica* Serovars: An *In Vitro* Study

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Salmonella enterica serovar Typhi and *Salmonella* Paratyphi are causative agents of enteric fever. *Salmonella* Typhi persists as a biofilm on gallstones. Hence, we studied the biofilm formation, antibiogram, and virulence genes of *S. enterica* serovars. Antibiogram of *S. enterica* serovars from human blood and stool samples were studied by Kirby–Bauer disk diffusion method and biofilm by microtiter plate method. We studied the minimum inhibitory concentration of the isolates by Vitek-2 semiautomated system. Polymerase chain reaction was done to detect *invA* and *spvC* genes. Of the 55 isolates studied, 36 (65.45%) were *Salmonella* Typhi, 13 (23.63%) were *Salmonella* Paratyphi A, 2 (3.64%) were *Salmonella* Typhimurium, and 4 (7.28%) were *Salmonella* spp. Resistance to ciprofloxacin and nalidixic acid were found to be 81.8% and 92.7%, respectively. Chloramphenicol and cotrimoxazole-susceptible strains were 98.18%. One each of *Salmonella* Typhi, *Salmonella* Paratyphi A, and *S. enterica* isolates formed weak biofilm at 28°C. However, at 37°C eight *Salmonella* Typhi produced weak biofilm in the presence of bile. One *Salmonella* Paratyphi A and two *Salmonella* spp. formed weak biofilm in the absence of bile. All the isolates had the *invA* gene. *Salmonella* Typhimurium had *invA* and *spvC* genes. Bile may contribute to biofilm formation and persistence of the *Salmonella* Typhi on gallstones, which may lead to carrier state. Changing antibiotic susceptibility pattern of *Salmonella* serovars is observed in our geographic area. The presence of *inv*A and *spvC* genes indicate the ability of invasiveness and intracellular survival.

Keywords: *Salmonella* Typhi, drug resistance, virulence genes, biofilm, PCR, India, *invA*, *spvC*, *Salmonella* Paratyphi A, antibiotic

Introduction

Salmonella enterica is a major bacterial agent that causes foodborne infection in humans all over the world. Many *S. enterica* serovars are present in the gut of the animals and released in the environment.[1] It causes zoonotic foodborne infection with serious impact on the economy of the society in the form of loss of productivity and increased medical expenses. *S. enterica* has more than 2,600 serovars, and important ones that cause human illness are *Salmonella* Typhi, *Salmonella* Paratyphi, *Salmonella* choleraesuis, *Salmonella* Infantis, *Salmonella* Enteritidis, and *Salmonella* Typhimurium.[2] Enteric fever is caused by *Salmonella* Typhi and *Salmonella* Paratyphi.[3] Nontyphoidal Salmonellae cause severe diarrhea and occasionally bacteremia.[1] It has become the most common cause of bacteremia in tropical Africa, with increasing incidence in young adults and HIV-infected individuals.[4]

Multiple drug-resistant (MDR) Salmonellae are resistant to three different classes of antimicrobial agents, namely ampicillin, trimethoprim–sulfamethoxazole, and chloramphenicol. Extensively drug-resistant *Salmonella* Typhi are MDR strains resistant to fluoroquinolones and third-generation cephalosporin.[5] During the 90s, there was increased isolation of MDR *S. enterica* serovars from different parts of India.[6] Few *Salmonella* Typhi and *Salmonella* Paratyphi A strains isolated in our laboratory from 2003 to 2006 were also MDR with re-emergence of sensitivity to chloramphenicol.[7] Ceftriaxone resistance is being reported from different parts of the world.[8,9] Moreover, *Salmonella* Typhi resistant to ceftriaxone, fluoroquinolones, and azithromycin are also reported from different parts of the world.[5]

Biofilm formation by *S. enterica* serovars plays a significant role in its pathogenicity.[10] Bile is known to induce the production of exopolysaccharides, which facilitates biofilm formation on gallstones. Biofilm helps the organisms to

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persist for a long time, leading to the carrier state.[11] Biofilm-forming *Salmonella* Typhi produce an increased amount of membrane matrix, which physically shields the bacteria, helps it to adhere to surfaces, and conserves energy in the stressful location. Conversely, the planktonic phenotype, produce flagella, show increased metabolic activity, and migration to form new colonies of infection[12]

Furthermore, the virulence of *S. enterica* serovars is due to the presence of several genes, of which the *inv*A gene is in Salmonella pathogenicity island and *spv*C gene is plasmid encoded. The gene *inv*A is responsible for the entry of the pathogen into the human epithelial cell and the gene *spvC* in intracellular survival.[13] However, the role of *spv*C in human infection is still controversial as this gene is sparsely detected in human *Salmonella* Typhimurium isolates, but not in *Salmonella* Typhi and *Salmonella* Paratyphi A.[14]

Human Salmonellosis are foodborne infections and studies on biofilm are mainly confined to food processing environments. It is known that the biofilm-forming capacity varies among different Salmonella strains and hence its persistence in food and food contact surfaces.[15] Thus, the strains of *Salmonella* causing human infections survive at both human body temperature (37°C) and at 28°C ± 2°C in food. Hence, we studied the biofilm-forming ability of clinical isolates of *S. enterica* serovars at two different temperatures in the presence and absence of bile. The biofilm-forming ability of the isolates were compared with the antibiotic susceptibility pattern and virulence genes like *invA* and *spvC* to know the association if any.

Materials and Methods

Isolation and identification of S. enterica *serovars*

S. enterica serovars ($n = 55$) isolated from clinical samples like blood and stool at the Department of Microbiology, from December 2016 to May 2018, were included in the study. This study has received approval from the Institutional Ethics Committee, KMC Mangalore. All the chemicals, media, and antibiotic discs for the study were procured from Hi-Media Laboratories Pvt. Ltd. (Mumbai, India).

BacT/ALERT blood culture bottles containing blood samples from patients with pyrexia of unknown origin was incubated aerobically into BacT/ALERT system (bioMérieux). Aliquots from positive blood culture bottles were subjected to Grams staining and subcultured onto chocolate agar, blood agar, and MacConkey's agar. Inoculated culture plates were incubated for 24 hr at 37°C.[16] Data regarding the presence or absence of gallstones in *Salmonella* culture-positive patients are not available.

Stool sample collected was inoculated into Selenite F broth, MacConkey's agar and incubated at 37°C. Also, a direct culture of the sample was done on MacConkey's agar. After 6 hr of incubation in Selenite F broth, it was subcultured onto MacConkey's agar. The plates were incubated aerobically at 37°C for 24 hr.[17]

Lactose nonfermenting colonies grown on MacConkey's medium were identified by biochemical reaction using VITEK-2 system (bioMérieux). Polyvalent (A-G) and monovalent antisera were used for serotyping of Salmonella spp. It was procured from the National Salmonella and Escherichia Center, Central Research Institute (Kasauli, India).

Antimicrobial susceptibility testing

Kirby–Bauer disk diffusion method was used to study the antibiotic susceptibility pattern of *Salmonella* spp. Two to three colonies of the biochemically confirmed Salmonella spp. were inoculated into Muller–Hinton broth, incubated for 4 h at 37°C, and the turbidity was adjusted to 0.5 McFarland standard. With the help of a sterile cotton swab, the broth culture was swabbed over the entire surface of the Muller Hinton agar (MHA) plates to get a lawn culture of the bacterium. Antibiotic discs of ampicillin, ceftriaxone, trimethoprim–sulfamethoxazole, ciprofloxacin, chloramphenicol, and nalidixic acid was placed on the MHA plates and incubated for 24 hr at 37°C.[18] Results were inferred as per Clinical and Laboratory Standards Institute (CLSI) guidelines as susceptible, intermediate, or resistant.[18] The quality control strain used was ATCC *Escherichia coli* 25922. Minimum inhibitory concentration (MIC) for ciprofloxacin and ceftriaxone was done by the VITEK-2 system (bioMérieux). For ciprofloxacin, isolates with MIC of ≤0.06 μg/mL is considered as sensitive, and ≥1 μg/mL as resistant. For ceftriaxone, isolates with MIC of ≤1 μg/mL is regarded as sensitive, >4 μg/mL as resistant.[18]

Detection of biofilm by crystal violet assay

S. enterica serovars were grown in Tryptic soy broth (TSB) for 18 hr at 37°C. Broth cultures were diluted (1:20) in saline, to match the turbidity of 0.5 McFarland standard. Two hundred microliters of diluted culture was added in triplicate into two sets of sterile round-bottomed plastic microtiter plates (Labtech Medico Pvt. Ltd., Kerala, India). *Enterococcus faecalis* ATCC 29212 was used as a positive control. TSB without inoculum served as negative control. One microtiter plate was incubated for 96 h at 28°C and the other at 37°C aerobically. The content of the wells was emptied, and washed with phosphate-buffered saline to remove the planktonic forms. The biofilm was fixed for 20 min by using methanol and then air dried. Biofilm was stained with 0.1% Crystal Violet solution for 20 min at room temperature and washed. Wells were dried, the dye was

Table 1. Primers Used in the Present Study

Primer	*Sequences 5′ to 3′*	*Gene*	*Annealing temperature (°C)*	*Product size (bp)*	*Reference*
*INV*A—F	ACAGTGCTCGTTTACGACCTGAAT	*inv*A	56	244	Chiu *et al.*[23]
*INV*A—R	AGACGACTGGTACTGATCGATAAT				
*SPV*C—F	ACTCCTTGCACAACCAAATGCGGA	*spv*C	56	571	Chiu *et al.*[23]
*SPV*C—F	TGTCTTCTGCATTTCGCCACCATCA				

resolubilized with 200 μL of 80% ethanol per well.[19] Absorbance was recorded at 550 nm in a microtiter plate reader (Multiskan™ FC Filter-based Microplate Photometer; Thermo Fisher Scientific).

MIC of bile (sodium deoxycholate) for different isolates and *Salmonella* Typhimurium ATCC 14028 was determined by micro broth dilution method. MIC of bile was 4% for 53 isolates and 5% for 2 of the isolates. One concentration below the MIC was taken as sublethal concentration. Hence, to check whether the sublethal concentration of bile has any effect on biofilm formation, TSB supplemented with 3% bile[20] was used, and the experiment was repeated as mentioned above. Three standard deviations (SDs) above the mean of the negative control were considered as cutoff (O.Dc). Based on the cutoff values, biofilm producers were classified as weak {O.Dc < O.D ≤ (2×O.Dc)}, moderate {(2×O.Dc) < O.D ≤ (4×O.Dc)}, strong {(4×O.Dc) < O.D}, and nonbiofilm producers {O.D ≤ O.Dc}.[21]

Detection of genes invA *and* spvC *by polymerase chain reaction*

Polymerase chain reaction (PCR) test was used to detect *invA* and *spvC* virulence genes in isolates of *S. enterica* serovars. DNA was extracted by boiling method; briefly, *Salmonella* strains were grown in 2 mL of TSB at 37°C for 18 h. The overnight cultures were centrifuged for 10 min at 10,000 *g*. Pellet obtained was washed with sterile distilled water twice and emulsified in 100 μL of nuclease-free water, heated for 10 min in a dry bath at 100°C to lyse the cells. Brief centrifugation of the lysate was done using Eppendorf Centrifuge 5430 (Eppendorf India Ltd., Chennai, India) and two microliters of the supernatant was used as DNA template.[22]

Primers were procured from Eurofins Genomics India Pvt. Ltd. (Bengaluru India). PCR was done by using the custom-synthesized primer pairs[23] as given in Table 1. Negative and positive controls were nuclease-free water and *Salmonella* Typhimurium ATCC 14028, respectively. Duplex PCR reaction mix contained 0.3 μmol/L of each primer, 200 μmol/L each of deoxynucleotide triphosphates, and 1.25 U of Taq polymerase with 1×buffer, and 25 mmol/L $MgCl_2$ and 1 μL DNA. The amplification reaction was done in a thermocycler (Bio-Rad, Inc.). Amplification condition included denaturation at 94°C for 1 min, followed by 30 cycles of 94°C for 30 sec, 56°C for 30 sec, 72°C for 2 min, and a final extension for 10 min at 72°C. The amplified product was resolved by 1.5% agarose gel electrophoresis using 1×Tris-acetate EDTA buffer and stained by 0.5 mg/mL ethidium bromide. Gel pictures were captured using the Gel documentation system (Alpha View 1.3.0; Alpha Innotech Corporation Multiimage Light Cabinet).[23]

Statistical analysis

Descriptive data are presented as frequencies and percentages. Chi-square test was applied for categorical data to compare the biofilm formation at different temperatures and antibiotic resistance. *p*-Value <0.05 was taken as significant. The statistical package SPSS ver11.0 for Windows (SPSS, Inc., Chicago, IL) was used for the statistical analysis.

TABLE 2. ANTIBIOTIC SUSCEPTIBILITY PATTERN OF *SALMONELLA ENTERICA* SEROVARS

	S. enterica *serovars* (n = 55)												Total (n = 55)		
	Salmonella *Typhi* (n = 36)			Salmonella *Paratyphi A* (n = 13)			Salmonella *Typhimurium* (n = 2)			Salmonella *spp.* (n = 4)					
Antibiotics tested	*S (%)*	*I (%)*	*R (%)*	*S (%)*	*I (%)*	*R (%)*	*S (%)*	*I (%)*	*R (%)*	*S (%)*	*I (%)*	*R (%)*	*S (%)*	*I (%)*	*R (%)*
Ampicillin	3 (97.3)	0 (0)	1 (2.7)	8 (61.5)	1 (7.7)	4 (30.8)	2 (100)	0 (0)	0 (0)	4 (100)	0 (0)	0 (0)	49 (89.1)	1 (1.8)	5 (9.1)
Cotrimoxazole	35 (97.3)	0 (0)	1 (2.7)	13 (100)	0 (0)	0 (0)	2 (100)	0 (0)	0 (0)	4 (100)	0 (0)	0 (0)	54 (98.2)	0 (0)	1 (1.8)
Chloramphenicol	35 (97.3)	0 (0)	1 (2.7)	13 (100)	0 (0)	0 (0)	2 (100)	0 (0)	0 (0)	4 (100)	0 (0)	0 (0)	54 (98.2)	0 (0)	1 (1.8)
Ciprofloxacin	2 (5.5)	0 (0)	34 (94.5)	1 (7.7)	1 (7.7)	11 (84.6)	2 (100)	0 (0)	0 (0)	3 (75.0)	0 (0)	1 (25.0)	9 (16.4)	1 (1.8)	45 (81.8)
Nalidixic acid	0 (0)	0 (0)	36 (100)	1 (7.7)	0 (0)	12 (92.3)	0 (0)	0 (0)	2 (100)	3 (75.0)	0 (0)	1 (25.0)	4 (7.3)	0 (0)	51 (92.7)
Ceftriaxone	36 (100)	0 (0)	0 (0)	13 (100)	0 (0)	0 (0)	2 (100)	0 (0)	0 (0)	4 (100)	0 (0)	0 (0)	55 (100)	0 (0)	0 (0)

I, intermediate; R, resistant; S, susceptible.

Results

Demographic data

Our study population included both male and female patients from whom 55 *S. enterica* serovars were isolated. The age of male patients varied from 13 to 80 years, and that of female patients from 12 to 69 years. The median age of male was 24 years, and that of the female was 24.5 years. Of the 55 isolates, 65.45% ($n=36$) were isolated from males and 34.55% ($n=19$) were isolated from females.

Salmonella *isolates included in the study*

In our study, 53 *S. enterica* isolates were recovered from blood and 2 from stool samples. Among these, 36 (67.92%) *Salmonella* Typhi, 13 (24.53%) *Salmonella* Paratyphi A, 2 (3.77%) each of *Salmonella* Typhimurium, and *Salmonella* spp. were isolated from blood. Two *Salmonella* spp. from stool and blood agglutinated with polyvalent *Salmonella* antiserum, but could not be serotyped further.

Antibiotic susceptibility of Salmonella *isolates*

Results of the antibiotic susceptibility test for the isolates by disk diffusion method are shown in Table 2. All the strains were susceptible to ceftriaxone and resistant to nalidixic acid by disk diffusion method. MIC for ciprofloxacin and ceftriaxone was done using the Vitek-2 semiautomated system. The MIC value for ciprofloxacin for the isolates ranged from 0.023 to 8 μg/mL. Strains showing MIC between >1 and 8 μg/mL for ciprofloxacin were considered as resistant. MIC of ceftriaxone ranged from 0.094 to 0.190 μg/mL and MIC was in the susceptible range. All the isolates were susceptible to ceftriaxone even by disk diffusion method.

Biofilm production

The ability to produce biofilm at two different temperatures (28°C and 37°C) in the presence and absence of bile was studied by microtiter plate method. The cutoff (O.Dc) value is taken as any value more than or equal to mean of negative control +3SD. One each of *Salmonella* Typhi, *Salmonella* Paratyphi A, and *Salmonella* spp. isolated from blood produced a weak biofilm (10.9%) in the presence and absence of bile at 28°C. At 37°C, eight *Salmonella* Typhi (14.54%) isolates from blood produced weak biofilm in the presence of bile. At the same temperature, in the absence of bile, only three (5.45%) isolates were able to form the biofilm, of which one was *Salmonella* Paratyphi A isolated from blood and the other two were *Salmonella* spp. isolated from stool. One of the weak biofilm-producing *Salmonella* Typhi isolates was susceptible to ceftriaxone and resistant to all other antibiotics tested. Rest of the weak biofilm-producing *Salmonella* Typhi and *Salmonella* Paratyphi A isolates were resistant to ciprofloxacin and nalidixic acid. However, the weak biofilm-producing stool isolates were sensitive to all the antibiotics tested. Results of biofilm production at 28°C and 37°C are depicted in Tables 3 and 4, respectively. The optical density values of the weak biofilm formers at two different temperatures along with their antibiotic susceptibility patterns are shown in Table 5.

Detection of virulence genes

PCR for the detection of *inv*A and *spv*C genes was performed on all isolates. All the isolates were positive for the *inv*A gene, and two strains of *Salmonella* Typhimurium had both *inv*A and *spv*C genes. The reault of PCR is depicted in Fig. 1.

Discussion

Salmonella Typhi was the predominant serovar isolated from blood followed by *Salmonella* Paratyphi A, as this region is endemic to typhoid and paratyphoid fever like many other states of India.[6,7] A retrospective study reported *Salmonella* Typhi (83.1%) as the predominant serovar isolated from blood followed by *Salmonella* Paratyphi A (16.9%).[24] Our isolation rate is at par with earlier Indian studies.

An increase in resistance to nalidixic acid (60.7%–93.8%), ciprofloxacin (13%–94%), and susceptibility to chloramphenicol was reported in India.[7,25,26] However, studies from other countries indicate a steep rise in MDR *Salmonella* strains ranging from 17.8% to 36.2% or more.[27–30] In the present study, ciprofloxacin and nalidixic acid-resistant strains were 81.8% and 92.7%, respectively. Moreover, 98.18% of the strains were susceptible to chloramphenicol and cotrimoxazole. All the isolates were susceptible to ceftriaxone by Vitek-2 system and disk diffusion method (Table 2). Thus, antibiogram of *Salmonella* serovars varies from one geographic area to the other. It is necessary to monitor the slow and continuous shifting pattern of drug resistance in each geographical area.

Table 3. Biofilm-Forming Ability of *Salmonella* spp. at 28°C in the Presence and Absence of Bile

	Salmonella enterica *serovars* (n=55)							
	Salmonella *Typhi* (n=*36*)		Salmonella *Paratyphi A* (n=*13*)		Salmonella *Typhimurium* (n=*2*)		Salmonella *spp.* (n=*4*)	
Type of biofilm	*+ Bile*	*− Bile*	*+ Bile*	*− Bile*	*+ Bile*	*− Bile*	*+ Bile*	*− Bile*
Weak	1	1	1	1	0	0	1	1
Moderate	0	0	0	0	0	0	0	0
Strong	0	0	0	0	0	0	0	0
No biofilm	35	35	12	12	2	2	3	3

p-Value 0.368; p-value <0.05 is significant.
+, Presence of bile; –, absence of bile.

TABLE 4. BIOFILM-FORMING ABILITY OF *SALMONELLA* SPP. AT 37°C IN THE PRESENCE AND ABSENCE OF BILE

	Salmonella enterica *serovars* (n = *55)*							
	Salmonella *Typhi* (n = *36)*		Salmonella *Paratyphi A* (n = *13)*		Salmonella *Typhimurium* (n = *2)*		Salmonella *spp.* (n = *4)*	
Type of biofilm	*+ Bile*	*− Bile*	*+ Bile*	*− Bile*	*+ Bile*	*− Bile*	*+ Bile*	*− Bile*
Weak	8	0	0	1	0	0	0	2
Moderate	0	0	0	0	0	0	0	0
Strong	0	0	0	0	0	0	0	0
No biofilm	28	36	13	12	2	2	4	2

p-Value 0.168; *p*-value <0.05 is significant.
+, Presence of bile; −, absence of bile.

Majority of the human Salmonella infections are foodborne.[15] Thus, the strains of salmonella causing human infections survive at human body temperature (37°C) and at 28°C ± 2°C in food, which made us to study the biofilm formation at two different temperatures. Subinhibitory concentration of bile is an inducer of biofilm at 37°C and regulates the expression of virulence genes.[20,21] An earlier study by Morium *et al.*[31] reports 100% of their clinical isolates (30 *Salmonella* Typhi and 11 *Salmonella* Paratyphi A) to be biofilm producers at 37°C. In the present study, only 22% ($n = 8$) of *Salmonella* Typhi produced weak biofilms in the presence of 3% bile at 37°C and 1 isolate at 28°C (Tables 3 and 4). This difference in the biofilm formation needs to be explored further, as it depends on the persistence of organism on gallstone, culture media, and environmental conditions.[32] In our study, all the *Salmonella* Typhi ($n = 36$) isolates were from patients suffering from enteric fever (active infection), and data regarding the presence of gallstone in these patients are not available. Moreover, bile at 3% concentration did not have a statistically significant effect on biofilm production, at both the temperatures in our experimental conditions probably because all were weak biofilm producers (Table 5). Earlier workers have shown that bile is an essential environmental signal sensed by *Salmonella* spp., at 37°C, and it plays a vital role in the regulation of multiple virulence-associated gene expression pathways.[33]

Walawalkar *et al.* studied the effect of bile-generated stress on *Salmonella* Typhi. They found that the isolates were able to adhere to the walls of the polystyrene wells by producing biofilms. Their study showed bile to induce persistent cell formation in *Salmonella* Typhi that are capable of biofilm formation and responsible for chronic infections/carrier state.[21] Since our *Salmonella* Typhi isolates were from active enteric fever cases they have formed only weak biofilm. However, follow-up of these patients after recovery for 6 months to know whether they go into a carrier state would have revealed whether bile induces persistent cell formation in *Salmonella* Typhi. Biofilm-forming *Salmonella* Typhi produces increased amount of membrane matrix, which physically shields the bacteria, helps it to adhere to surfaces, and conserves energy in the stressful location.[12] Although the isolates in our study have produced weak biofilm in the presence and absence of bile at two

TABLE 5. WEAK BIOFILM-PRODUCING *SALMONELLA* ISOLATES AT TWO DIFFERENT TEMPERATURES WITH OPTICAL DENSITY VALUES AND THEIR ANTIBIOTIC SUSCEPTIBILITY PATTERN

		Biofilm at 37°C		*Biofilm at 28°C*		*Antibiotic name*					
		OD values		*OD values*							
Samples tested	*Isolates*	*+ Bile (cov 0.582)*	*− Bile (cov 0.115)*	*+ Bile (cov 0.590)*	*− Bile (cov 0.136)*	*AMP*	*COT*	*C*	*CIP*	*NA*	*CTR*
Blood	*Salmonella* Typhi	**0.614**	0.085	0.341	0.100	S	S	S	R	R	S
Blood	*Salmonella* Typhi	**0.690**	0.084	0.314	0.107	S	S	S	R	R	S
Blood	*Salmonella* Typhi	**0.635**	0.082	0.196	0.111	S	S	S	R	R	S
Blood	*Salmonella* Typhi	**0.793**	0.08	0.397	0.118	R	R	R	R	R	S
Blood	*Salmonella* Typhi	**0.624**	0.083	0.352	0.119	S	S	S	R	R	S
Blood	*Salmonella* Typhi	**0.890**	0.087	0.35	0.118	S	S	S	R	R	S
Blood	*Salmonella* Typhi	**0.620**	0.083	0.281	0.127	S	S	S	R	R	S
Blood	*Salmonella* Typhi	**0.583**	0.085	0.313	0.102	S	S	S	R	R	S
Blood	*Salmonella* Typhi	0.518	0.104	**0.665**	**0.139**	S	S	S	R	R	S
Blood	*Salmonella* Paratyphi A	0.333	**0.141**	**0.611**	**0.155**	S	S	S	R	R	S
Blood	*Salmonella* spp.	0.162	0.095	**0.601**	**0.137**	S	S	S	S	S	S
Stool	*Salmonella* spp.	0.354	**0.142**	0.377	0.115	S	S	S	S	S	S
Stool	*Salmonella* spp.	0.273	**0.131**	0.313	0.128	S	S	S	S	S	S

OD values in bold indicate weak biofilm producers.
AMP, ampicillin; C, chloramphenicol; CIP, ciprofloxacin; COT, cotrimoxazole; Cov, cut of value; CTR, ceftriaxone; NA, nalidixic acid; OD, optical density.

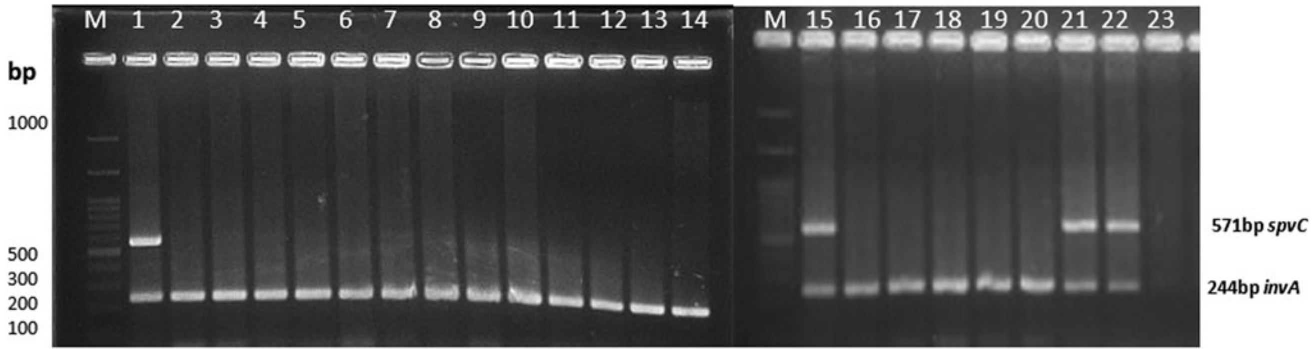

FIG. 1. Agarose gel picture of Duplex PCR for the detection of *spvC* and *invA* gene in *Salmonella enterica* serovars. *Lanes 1 and 15:* Positive control; *lanes 1–6: Salmonella enterica* ser Typhi. *Lanes 7–14: Salmonella enterica* ser Paratyphi A; *lanes 16–20: Salmonella enterica* serovars; *lanes 21* and *22: Salmonella enterica* ser Typhimurium; *lane 23:* Negative control. M, molecular weight marker; PCR, polymerase chain reaction.

different temperatures, this is a foundation for further studies to uncover the mechanism of biofilm formation in *Salmonella* Typhi and *Salmonella* Paratyphi A to discover novel genes associated with the development of the relapse and carrier state in human infections.

In our study, no significant association was seen between the ability to form biofilms and antibiotic resistance (Table 5). However, a similar study by Raza *et al.* showed that 40% of strong biofilm-producing, drug-resistant *Salmonella* Typhi was excreted in the stool for a long duration, thus acting as a source of infection in the community.[32] They showed that bile does not alter the drug resistance profile of bacteria, rather they protect the bacteria from the effect of drug. Therefore, bile may be responsible for the formation of biofilm on gallstones, which may lead to the persistence of *Salmonella* Typhi and thus the typhoid carrier state and prolonged excretion in stool. Furthermore, *in vitro* studies are needed to explore the ability of *Salmonella* spp. to form biofilm on gallstones in the presence and absence of bile to reveal the mechanism of colonization in gallbladder by these organisms.

In *Salmonella* serovars, genes *invA* and *spvC* are located on the chromosome and plasmid, respectively. Genes *invA* and *spvC* play roles in penetration into the epithelial cells and intracellular survival of bacteria.[2] In our study, all the isolates were positive for the *inv*A gene, and *Salmonella* Typhimurium isolates were positive for the *spv*C gene (Fig. 1). An earlier study reports the absence of *inv*A and *spv*C genes in 6.3% and 73% isolates of *Salmonella* spp., respectively. Their research also indicates a very low prevalence of the *spvC* gene in human isolates.[14,34] Our findings are in line with the earlier workers. However, the role of *spv* C in human infection is controversial as this gene is sparsely detected in *Salmonella* Typhimurium causing human infection, but not in human host-adopted *Salmonella* Typhi and *Salmonella* Paratyphi A.[13] In our study, there was no significant association between the occurrence of *inv*A and or *spv*C gene and the ability to form biofilms or drug resistance. Unfortunately, there are no published reports to compare the results of our study. Moreover, the number of salmonellae that produced biofilm were few to deduce any significance.

Our study has some limitations. Clinical data of the enteric fever patients, such as carrier state/cholecystitis/cholecystectomy, were not available. There are a lot of other genes involved in Salmonella pathogenesis. Due to the short duration of the study we could detect only two of the virulence genes. The molecular mechanism involved in biofilm formation and persistence of bacteria on gallstone needs to be explored further.

To conclude, *Salmonella* Typhi and *Salmonella* Paratyphi A were the predominant serovars identified from blood samples. More than 80% of the isolates were resistant to fluoroquinolones, and all were susceptible to ceftriaxone. More number of strains were sensitive to chloramphenicol and cotrimoxazole. Monitoring of antimicrobial resistance is necessary to detect the slow and continuous changing pattern of drug resistance. No association was found between biofilm formation and drug resistance in the study setting. Bile may contribute to biofilm formation and persistence of the *Salmonella* Typhi on gallstones, which may lead to carrier state.

Author Contributions

B.D. designed the experiment; D.K. performed tests; B.D. and D.K. analyzed the data; D.K. and B.D. wrote the article and approved the final version.

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Ethics Approval

This study has been approved by the Institutional Ethics Committee, Kasturba Medical College, Mangalore, Manipal Academy of Higher Education, Manipal [Reference No: IEC KMC MLR 10-17/184].

Availability of Supporting Data

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Disclosure Statement

No competing financial interests exist.

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